

Product Requirements Document (PRD)

AssetFlow AI

AI-Powered Smart Asset Lifecycle Management Platform

Version: 1.0

Prepared For: Hackathon

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1. Executive Summary

Project Name

****AssetFlow AI****

Vision

AssetFlow AI is an intelligent Asset Lifecycle Management platform that enables organizations to efficiently manage physical assets from procurement to disposal.

The platform combines:

- Asset Management
- QR Code Tracking
- AI Assistant
- Predictive Maintenance
- Analytics
- Smart Automation

The goal is to reduce operational costs while increasing asset visibility and utilization.

2. Problem Statement

Many organizations still manage their assets using Excel sheets or paper records.

This creates several issues:

- Assets are frequently lost.
- Duplicate purchases happen due to lack of visibility.
- Maintenance is reactive instead of proactive.
- Warranty periods expire unnoticed.

- Audits become difficult.
- Employees do not know who owns an asset.
- Management cannot analyze asset performance.

Current enterprise asset management software is often expensive and difficult to customize.

3. Proposed Solution

AssetFlow AI centralizes all asset information into one intelligent platform.

The application enables organizations to:

- Register assets
- Generate QR Codes
- Assign assets to employees
- Track movement
- Manage maintenance
- Predict failures using AI
- Search assets using natural language
- Generate reports
- Monitor utilization

Every asset has a complete digital lifecycle.

4. Target Users

4.1 Administrator

Responsibilities

- Manage assets
- Manage users
- Assign assets
- View reports
- Configure departments

Pain Points

- Manual records
- Difficult audits
- Lost assets

4.2 Employee

Responsibilities

- View assigned assets
- Report problems
- Scan QR code
- Request replacement

Pain Points

- Slow maintenance
- No ownership history

4.3 Technician

Responsibilities

- Receive maintenance tickets
- Update repair status
- Maintain repair history

Pain Points

- No repair records
- Difficult scheduling

4.4 Manager

Responsibilities

- View department assets
- Approve requests
- Analyze reports

Pain Points

- Budget planning
- Low asset utilization

5. Product Goals

The system should:

- Reduce asset loss
- Improve utilization
- Automate maintenance
- Simplify audits
- Reduce manual work
- Improve transparency
- Support AI-based decision making

6. Success Metrics

Operational Metrics

- 95% tracking accuracy
- 40% faster asset search
- 30% reduction in maintenance delays

Business Metrics

- Lower maintenance cost
- Fewer duplicate purchases
- Better asset utilization

User Metrics

- Daily Active Users
- Number of QR scans
- Average maintenance completion time

7. Scope

Included in MVP

Authentication

- Login
- Logout
- Forgot Password
- JWT Authentication

Asset Management

- Add Asset
- Edit Asset
- Delete Asset
- Search Asset
- Filter Asset
- Asset Details

QR Management

- Generate QR
- Scan QR
- Open Asset Profile

Assignment

- Assign Asset
- Return Asset
- Transfer Asset

Maintenance

- Raise Ticket
- Technician Assignment
- Repair Status
- Maintenance History

Dashboard

- Total Assets
- Available Assets
- Assigned Assets
- Under Repair
- Warranty Expiry

- Maintenance Due

Reports

- Department Report
- Vendor Report
- Utilization Report
- Maintenance Report

AI Features

- AI Chat Assistant
- Smart Search
- Predictive Maintenance
- OCR Invoice Scanner

8. Functional Requirements

Admin

Can

- Add assets
- Edit assets
- Delete assets

- Manage users
- Generate reports
- Assign assets

Employee

Can

- View assigned assets
- Raise maintenance request
- Scan QR
- Return asset

Technician

Can

- View assigned tickets
- Update repair status
- Upload repair notes

Manager

Can

- View dashboards
- Approve requests
- Export reports

9. Non-Functional Requirements

- Responsive UI
- Fast loading (<2 seconds)
- Secure authentication
- Role-based authorization
- Audit logs
- Scalable architecture
- Mobile-friendly
- Cloud deployment
- Backup support

10. User Journey

```text

Login



Dashboard



Add Asset



Generate QR



Assign Employee  
↓  
Employee Uses Asset  
↓  
Maintenance Required  
↓  
Technician Repairs  
↓  
Manager Views Analytics  
↓  
Asset Disposal  
...

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# 11. MVP Features

| Feature                | Priority     |
|------------------------|--------------|
| -----                  | -----        |
| Login                  | Must Have    |
| Dashboard              | Must Have    |
| Asset CRUD             | Must Have    |
| QR Generation          | Must Have    |
| Asset Assignment       | Must Have    |
| Maintenance            | Must Have    |
| Reports                | Must Have    |
| AI Chat                | Should Have  |
| OCR Scanner            | Should Have  |
| Predictive Maintenance | Nice to Have |

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## # 12. Future Scope

- RFID Tracking
- IoT Sensors
- GPS Tracking
- Mobile App
- Voice Assistant
- AI Damage Detection
- Blockchain Asset Ledger
- ERP Integration
- Multi-Tenant Support

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## # Product Vision

> Build the smartest AI-powered Asset Lifecycle Management Platform that helps organizations know  
\*\*what they own, where it is, who is using it, when it needs maintenance, and when it should be replaced.\*\*